OB+RCT
CATE/Overlap/High dimension/Matching

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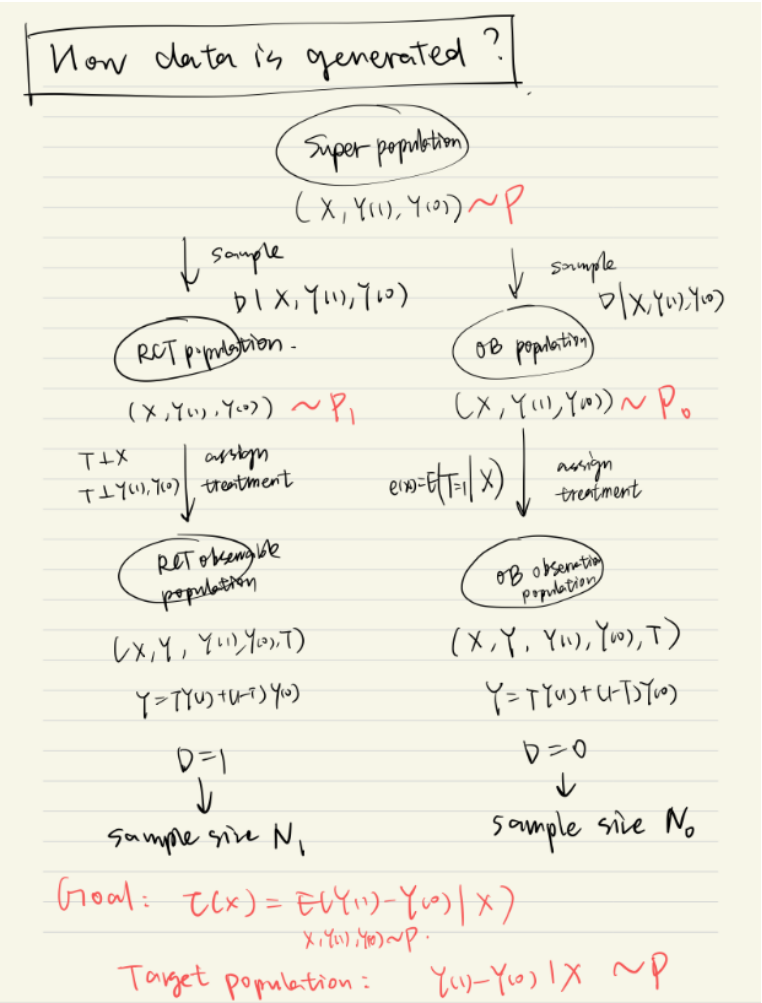
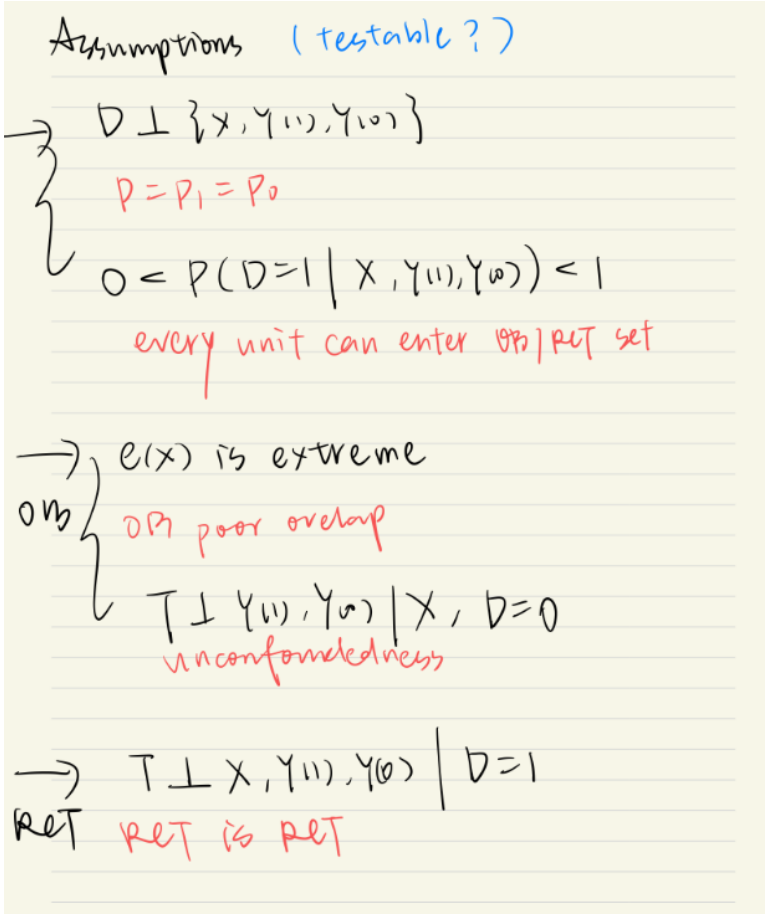
combine rct and ob - why?

- RCT+EC
  - RCT so small
  - RCT may only cover some specific patient(X support region)
- OB+RCT
  - OB with poor overlap(in what sense?) our case
  - OB with unmeasured confounding

combine rct and ob - a data generating view

In super population framework, both RCT and OB are samples from a stochastic distribution. (not finite population and design-based experiment framework)





OB - overlap

- Overlap condition influences the performance of estimators (Busso, DiNardo, and McCrary 2014)
- Theoretically, Overlap assumption is a must for the identification of ATE and Strict Overlap is needed for statistical analysis in many papers. (e.g. Causal RF)

? (Crump et al. 2009) [ATE] Characterize(by ps score) optimal subsamples for which the average treatment effect can be estimated most precisely.

we also consider the problem without imposing this restriction. Formally, we choose an estimand $\tau_{S,\mathbb{A}}$ by choosing the set $\mathbb{A} \subset \mathbb{X}$ that minimizes

$$V_{\mathbb{A}}^{\text{eff}} = \frac{1}{\{E(1_{X\in\mathbb{A}})\}^2} E\left[1_{X\in\mathbb{A}}\left\{\frac{\sigma_1^2(X)}{e(X)} + \frac{\sigma_0^2(X)}{1-e(X)}\right\}\right].$$

Defining $q(\mathbb{A}) = \text{pr}(X \in \mathbb{A}) = E(1_{X\in\mathbb{A}})$, we can write the objective function as

$$V_{\mathbb{A}}^{\text{eff}} = \frac{1}{q(\mathbb{A})} E\left\{\frac{\sigma_1^2(X)}{e(X)} + \frac{\sigma_0^2(X)}{1-e(X)} \middle| X \in \mathbb{A}\right\}.$$

We seek $\mathbb{A} = \mathbb{A}^*$, which minimizes $V_{\mathbb{A}}^{\text{eff}}$ among all closed subsets $\mathbb{A} \subset \mathbb{X}$.

- (Li, Thomas, and Li 2019) [ATE] Overlap Weighting

OB - overlap

- why poor overlap?
  -
- poor overlap, then?
- when does that matter

OB - overlap & balance

- overlap and balance

$$xx$$

- high-dimensional: (1) many covariates; (2) complex interactions among covariates in outcome model or propensity model
- Strict overlap is very 'strict' in high-dimensional setting. (D'Amour et al. 2021)[1]
- From data-generating view, if we have high-interaction confounding, then poor high-dimension overlap [?]

[1] Strict overlap restricts global discrepancies between the covariate distributions in the treated and control populations. These restrictions become more restrictive as the dimension grows large.

Overlap in CRF - high-dimensional/heterogeneity

An example of strict overlap in high-dimension case.

- Bias of estimation(Wager and Athey 2018)

$$|\mathbb{E}[\hat{\tau}(x)] - \tau(x)| \lesssim 2Md\left(\frac{\eta s}{2k-1}\right)^{-\frac{1}{2}\frac{\log((1-\alpha)^{-1})}{\log(\alpha^{-1})}\frac{\pi}{d}} = \mathcal{O}\left(s^{-\frac{1}{2}\frac{\log((1-\alpha)^{-1})}{\log(\alpha^{-1})}\frac{\pi}{d}}\right)$$

  - bounded covariates $X_1, \ldots X_s \in [0,1]^d$
  - M-bounded and Lipschitz continuous conditional mean functions
  - strict overlap: for some $\eta > 0$ and all $x \in [0,1]^d$, $\eta < e(x) < 1 - \eta$.
  - A tree predictor grown by recursive partitioning is α-regular
- Tree beats NN under settings of low-dimensional signal structure[2]
- More
  - is strict overlap plausible for tree? how does the balance of data influence the structure(then the performance) of the tree?
  - could match/balance based weighting be used in CRF?[3]

[2] a data-driven way to determine which nearby observations receive more weight

[3] "An important challenge for future work is to design splitting rules that can automatically choose which characteristic of the training data to split on."

Overlap in PLM and generalized FWL theorem

Consider Partial linear model

$$E(Y \mid X, Z) = \beta^T Z + \theta(X)$$

i.e.

$$Y - E(Y|X) = \beta(Z - E(Z|X)) + U, \; E(U|X, Z) = 0$$

In causal settings, given unconfoundedness, $E(Y(1)|X) = \beta + \theta(X)$, $E(Y(0)|X) = \theta(X)$, ATE = β(our estimand).

- (Robinson 1988) regression of residuals estimator $\hat{\beta} = S^{-1}_{Z-\hat{Z}} S_{Z-\hat{Z}, Y-\hat{Y}}$ is consistent $\sqrt{N}(\hat{\beta} - \beta) \xrightarrow{d} N(0, \sigma^2 \Phi^{-1})$. if
  - $\Phi = E[(Z - E(Z|X))(Z - E(Z|X))^\top]$ is positive definite.
    - $\Phi^{-1} = \frac{1}{E(e(x)(1-e(x))} = \frac{1}{E(Var(Z|X))}$ (convergence rate & overlap)
  - kernel estimators for first step, other ..

Overlap in R-learner

Considering HTE, now the PLM model becomes:

$$E(Y \mid X, Z) = \beta(X)Z + \theta(X)$$

i.e.

$$Y - E(Y|X) = \beta(X)(Z - E(Z|X)) + U,\ E(U|X, Z) = 0$$

$$Y - m(X) = \beta(X)(Z - e(X)) + U$$

In causal settings, given unconfoundedness, $E(Y(1)|X) = \beta(X) + \theta(X)$, $E(Y(0)|X) = \theta(X)$, CATE $= \beta(X)$ (our estimand).

- (Nie and Wager 2021) equivalently

$$\tau^*(\cdot) = \arg\min_{\tau} \mathbb{E}\left[\left(Y - m^*(X) - (W - e^*(X))\tau(X)\right)^2\right]$$

  therefore estimator(minimize R-loss, cross fitting):

$$\hat{\tau} = \arg\min_{\tau} \left(\hat{L}_n(\tau) + \Lambda_n(\tau)\right)$$

  - penalized kernel R-learner is near oracle(depends on the local robustness property of R-loss) if
    - strict overlap: turning regret bound into MSE bound

$$(1-\eta)^{-2}R(\tau) < E[\{\tau(X_i) - \tau^*(X_i)\}^2] < \eta^{-2}R(\tau)$$

    - RKHS, good estimate of nuisance functions, kernel, ...

OB - define/calculate overlap

$e(x)(1 - e(x))$

$\frac{1}{e(x)(1-e(x))}$

local 'chengdu' of overlap

$E(e(x)(1 - e(x)))$

OB - define/calculate balance

Need a metric to describe the distance of two distributions.

- one-dimensional norm(e,g. overlap region of propensity score[4]); mean-matching/pair-matching (Rubin 1973a)
- Mahalanobis distance[5] (which also contain terms that are functions of the covariates like power/interactions)
- p-values form KS test and paired t-tests (general because the same scale) (Diamond and Sekhon 2013)
- combine ps-score and original X and ..(better firstly orthogonalize by regressing)

[4] for ps score, better to use the linear predictor because it is note close to 0 or 1 (Diamond and Sekhon 2013)

[5] MD does not perform well when covariates have nonellipsoidal distributions?

OB - matching

- (Rubin 1973b) [OB, ATT, one-dimension X, homogeneous/parallel effect]. Use percent reduction in bias as a metric. Matching and regression can reduce bias in observational studies.
- (Diamond and Sekhon 2013) The GenMatch algorithm searches a range of distance metrics to find the particular measure that optimizes post-matching covariate balance. iteratively checking the specification of the propensity score model.

combine rct and ob

- (Liu et al. 2025) Introduce a double-indexed notation for potential outcomes to define causal estimands transparently and clarify distinct sources of implicit bias. A_i treatment status; Z_i data source.

$$Y_i = Y_i^1(1)A_iZ_i + Y_i^1(0)(1-A_i)Z_i + Y_i^0(1)A_i(1-Z_i) + Y_i^0(0)(1-A_i)(1-Z_i)$$

$$b(X_i) = E(Y_i \mid Z_i = 1, A_i = 0, X_i) - E(Y_i \mid Z_i = 0, A_i = 0, X_i)$$

- (Kim et al. 2025) Define weak overlap; covariate shift between sources.

Simulations settings

Under what kind of settings would our methods be better than other? N obs data, M rct data; d covariates, within which c are confounders and $d - c$ are prognsotic covariates(assume no instrumental variables)

- N/M
- $e(x)$, overlap, well-specified or not
- $\mu_0(x), \mu_1(x)$, the response surfaces and CATE $\tau(x)$
- c/d, how many irrelevant covariates
- high-dimensional or not (d/N or c/N, complexity of models)

Simulations settings - how to simulate X

Types of X: xx
(D'Amour et al. 2021) Covariate sequence X is a stochastic process $(X_{(k)})_{(k>0)}$, for any single problem, the investigator selects a finite set of covariates $X_{1:p}$ from the finite pool of covariates $(X_{(k)})_{(k>0)}$.

Simulations settings - linearity of outcome model

If the true outcome model is linear, $\mu_{(w)}(x) = x\beta(w)$, then

- the difference of them, i.e. CATE, is linear in X
- fitting one model for each z is equivalent to adding an interaction term Xz
- achieving sample balance[6] just involves balancing the X_i
- If the response surface is nonlinear, all estimates are general biased even if $\bar{x}_1 = \bar{x}_2$. (Rubin 1973b) [ATT, one-dimension]

[6] $\frac{1}{n}\sum_{i=1}^{n}\frac{Z_i\phi_j(X_i)}{e(X_i)} = \frac{1}{n}\sum_{i=1}^{n}\frac{(1-Z_i)\phi_j(X_i)}{1-e(X_i)} = \frac{1}{n}\sum_{i=1}^{n}\phi_j(X_i)$

Some thoughts and questions

- why does overlap influence the performance of estimators in math[7]
- What is target population? After matching/weighting, would that actually change? (can visualize X)
- After matching/weighting, is the balance better?
- metrics of CATE estimation for simulation? [8]
- data-augmenting then modeling v.s weighting in DL/Tree models (would it be different if we double one sample or weight it by two in model?)
- poor overlap in high-order-interaction-terms sense? hard to grow a tree? [how much overlap, i.e. how strong tails of $e(x)$]

[7] 'If the imbalance of the covariates between the two groups is large, the model-based results heavily relies on extrapolation in the region with little overlap, which is sensitive to the model specification assumption.'

[8] (1) $\frac{1}{n}\sum_{i=1}^{n}(\hat{\tau}_i - \tau_i)^2$; (2) $\frac{1}{n}\sum_{i=1}^{n}\hat{\tau}(x_i) - \frac{1}{n}\sum_{i=1}^{n}\tau(x_i)$; (3) $E(\hat{\beta} - \beta)^2$

Summary

- Goal: better estimate CATE
  - data: large-scale Post-OB data and small-scale RCT data
  - problem: poor overlap and severe unbalance
- Method: matching-based
  - data augmentation
  - ITE-imputing loss
- Data source: simulation and real-world data
- Further
  - when would the methods work better(the choice of caliper for matching, the size of ob&rct, the degree of non-overlap, the use of previous algorithm or not, etc)
  - an end-to-end rather than step-wise algorithm
  - multi-treatment

references

Busso, Matias, John DiNardo, and Justin McCrary (2014). "New evidence on the finite sample properties of propensity score reweighting and matching estimators". In: *Review of Economics and Statistics* 96.5, pp. 885–897.

Crump, Richard K et al. (2009). "Dealing with limited overlap in estimation of average treatment effects". In: *Biometrika* 96.1, pp. 187–199.

D'Amour, Alexander et al. (2021). "Overlap in observational studies with high-dimensional covariates". In: *Journal of Econometrics* 221.2, pp. 644–654.

Diamond, Alexis and Jasjeet S Sekhon (2013). "Genetic matching for estimating causal effects: A general multivariate matching method for achieving balance in observational studies". In: *Review of Economics and Statistics* 95.3, pp. 932–945.

Kim, Seok-Jin et al. (2025). "Transfer learning of cate with kernel ridge regression". In: *arXiv preprint arXiv:2502.11331*.

Li, Fan, Laine E Thomas, and Fan Li (2019). "Addressing extreme propensity scores via the overlap weights". In: *American journal of epidemiology* 188.1, pp. 250–257.

Liu, Bo et al. (2025). "Leveraging External Controls in Clinical Trials: Estimands, Estimation, Assumptions". In: *arXiv preprint arXiv:2503.21081*.

Nie, Xinkun and Stefan Wager (2021). "Quasi-oracle estimation of heterogeneous treatment effects". In: *Biometrika* 108.2, pp. 299–319.

Robinson, Peter M (1988). "Root-N-consistent semiparametric